



BEST PRACTICE ENGINEERING CONTROL GUIDELINES TO

Control Worker Exposure to Respirable Crystalline Silica during Asphalt Pavement Milling





DEPARTMENT OF HEALTH AND HUMAN SERVICES
Centers for Disease Control and Prevention
National Institute for Occupational Safety and Health

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Volvo Construction Equipment, and Roadtec Inc.

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Executive Summary

This document represents more than ten years of collaborative research by labor, industry, and government to reduce respirable crystalline silica exposure during asphalt pavement milling in highway construction. The collaborative research began when the Silica/Asphalt Milling Machine Partnership was formed at the 2003 National Asphalt Pavement Association (NAPA) Annual Meeting, and studies on milling machine dust controls began later that year. The Silica/Asphalt Milling Machine Partnership is coordinated by NAPA and includes all U.S. and foreign manufacturers of heavy construction equipment that currently sell pavement-milling machines to the U.S. market. In addition to NAPA and the equipment manufacturers, the Silica/Asphalt Milling Machine Partnership includes numerous paving contractors, the International Union of Operating Engineers, the Laborers International Union of North America, the Association of Equipment Manufacturers, and government organizations including the Occupational Safety and Health Administration (OSHA), the Federal Highway Administration, and the Centers for Disease Control and Prevention's (CDC's) National Institute for Occupational Safety and Health (NIOSH).

Based on the research of the Silica/Asphalt Milling Machine Partnership, NIOSH recommend engineering controls that include ventilation controls in addition to water-sprays used to cool the cutting teeth. Milling machines that adopt this well-designed dust control approach have been shown to control worker exposures to respirable crystalline silica from asphalt milling operations during 21 days of personal breathing zone sampling at 11 different highway construction sites [NIOSH 2013b,c]. The 42 personal breathing zone air sampling results (21 days with 2 workers per day) were all below the NIOSH recommended exposure limit of 0.05 mg/m^3 for respirable crystalline silica ranging from below the limit of detection up to 0.024 mg/m^3.

NIOSH and the Silica/Asphalt Milling Machine Partnership recommend ventilation controls be placed on all new half-lane and larger asphalt milling machines to reduce worker exposure to respirable crystalline silica. It is also recommended that water-sprays be used to suppress dust on milling machines if ventilation dust controls are not available. The ventilation and water-spray dust controls described in this document represent the current knowledge of best practices developed by the Silica/Asphalt Milling Machine Partnership for controlling dust during asphalt pavement milling. This document should not limit manufacturers from implementing future improvements in dust control as new technologies or inventions become available. Manufacturers can use the procedures in Appendices A and B of this document to evaluate current and future dust controls.

Authors and Acknowledgments

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The Silica/Asphalt Milling Machine Partnership

The guidelines presented in this document are a result of a collaborative effort by labor, industry, and government to reduce respirable crystalline silica exposure during asphalt pavement milling in highway construction. This Silica/Asphalt Milling Machine Partnership (Partnership) is coordinated by the National Asphalt Pavement Association (NAPA) and includes all U.S. and foreign manufacturers of heavy construction equipment that currently sell pavement-milling machines to the U.S. market. In addition to NAPA and the equipment manufacturers, the Partnership includes numerous paving contractors, the International Union of Operating Engineers, the Laborers International Union of North America, the Association of Equipment Manufacturers, and government organizations including the Occupational Safety and Health Administration (OSHA), the Federal Highway Administration, and the Centers for Disease Control and Prevention's (CDC's) National Institute for Occupational Safety and Health (NIOSH).

Many of the partners are the same members of a previous partnership to control asphalt fume exposures during paving. The Asphalt Paving Partnership was a finalist in the 1998 Innovation in American Government Award, presented by the Ford Foundation and Harvard University's Kennedy School of Government. The Asphalt Paving Partnership was responsible for successfully implementing engineering controls on all highway-class pavers to reduce exposure to asphalt fumes [NIOSH 1997]. The same partnership strategy is being used to implement engineering controls on pavement milling machines to reduce silica exposures. The aim of the Silica/Asphalt Milling Machine Partnership, and the focus of this document, is to identify engineering controls that could be used to reduce silica exposures among workers on half-lane and larger asphalt milling machines. The engineering controls include ventilation controls in addition to the water-spray systems used on the machines for cooling the cutting teeth. The recommendations developed in this report are based on dust control studies conducted on asphalt pavement surfaces. NIOSH and the Partnership have not evaluated the effectiveness of the best practice dust controls on other non-asphalt road surfaces such as concrete.

Five manufacturers of half-lane and larger cold-milling machines developed silica dust control strategies for their respective milling machines. NIOSH and the Partnership conducted laboratory and field studies to evaluate these controls for their effectiveness at preventing worker exposures to respirable crystalline silica. NIOSH recommend ventilation controls in addition to the water-spray systems used for cooling cutting teeth. An effective water-spray application system can also provide dust suppression at the source of dust generation on asphalt milling machines that are not fitted with ventilation controls. Ventilation controls on milling machines provided dust control capable of reducing worker exposures below the NIOSH recommended exposure limit (REL) of 0.05 mg/m^3 for respirable crystalline silica. Results indicated that these engineering controls can protect workers on asphalt milling machines from silica exposures.

In April of 2002, NIOSH met with the Association of Equipment Manufacturer's Compaction and Paving Machinery Technical Committee to propose the Silica/Asphalt Milling Machine Partnership. In June 2002, NIOSH met with representatives from NAPA, the Asphalt Institute, and the Harvard School of Public Health to propose the Silica/Asphalt Milling Machine Partnership. The Partnership was formed at the 2003 NAPA Annual Meeting, and studies on milling machine dust controls began later that year.

Milling-machine manufacturers that are Partnership members worked with research engineers from the NIOSH Division of Applied Research and Technology and the NIOSH Office of Mine Safety and Health Research to implement and test prototype dust emission-control systems based in part upon the transfer of technology from similar underground mining machinery. From 2003 to 2006, the Partnership helped facilitate six in-depth field surveys during road milling jobs. The first milling-machine field survey served as a pilot study for evaluating the effectiveness of water spray controls for a cold milling machine [NIOSH 2004]. Other field surveys provided data on the effectiveness of water spray controls for one state-of-the-art milling machine model from each of the four largest domestic manufacturers [NIOSH 2007a; 2009a,b,c].

The results of data analyses performed in 2007 suggested that improvements in engineering dust-emission controls were needed. Consequently, additional field work to evaluate the new designs also was needed. A controlled test site was located in 2008 with Silica/Asphalt Milling Machine Partnership assistance and utilized for the testing. During 2009, the data from this testing were evaluated, and two techniques were found to appreciably and statistically significantly reduce respirable dust levels in and around the milling machine [NIOSH 2011d]. These were ventilation of the cutter housing and primary conveyor areas and the addition of water sprays in the transfer area from the cutter housing to the primary conveyor loading area, with spray orientation counter to material flow. NIOSH and Partnership members concluded that further optimization of the dust emission-control systems followed by additional field testing should occur.

Another large field study was conducted in 2010 and found that the same water-spray dust-suppression controls that were tested in 2008 did not show the same reductions in respirable dust [NIOSH 2011c]. However, the ventilation control combined with water-sprays used to cool cutting teeth showed statistically significant reductions in respirable dust during both 2008 and 2010, and these results were more consistent than the results from any control that used water-spray dust suppression alone.

During 2011–2012, four laboratory tracer gas studies were conducted for four different manufacturers [NIOSH 2011a,b; 2012; 2013a]. The laboratory tracer gas studies were conducted at milling machine manufacturing facilities or milling contractor-owned facilities to determine the capture efficiency of the ventilation control before shipping the machine to a test location. The results from the laboratory studies indicated that tracer gas capture efficiencies of the evaluated ventilation controls were generally greater than 90%.

Although the earlier Silica/Asphalt Milling Machine Partnership studies did not show consistent reductions in respirable dust concentrations using only water-spray systems, one manufacturer requested additional testing of a wet drum design in 2012. The wet drum design uses gravity to release water from inside the drum through nozzles to the cutting surface to suppress dust. The results of this study showed that, while the wet drum appeared to provide increased control over the baseline water-spray configuration, the two were not statistically significantly different [NIOSH 2013d]. In addition, the magnitude of the exposure reduction did not appear to be sufficient to protect workers from excessive crystalline silica exposures. Based on the results of the wet drum evaluation and previous studies using only water-spray dust suppression, NIOSH recommended additional testing of ventilation-based controls in addition to the existing water-spray configurations.

In the summer and fall of 2012, NIOSH and the Partnership began conducting field tests of milling machines fitted with ventilation controls that had been developed by each manufacturer. The field testing consisted of full-shift personal breathing zone samples for respirable crystalline silica. Breathing zone samples were collected for the operator and the ground person who were performing the normal employee work activities of asphalt pavement milling on highway construction sites. The goal of the testing was to complete the worker exposure assessment for at least 9 days over a minimum of 3 sites for each manufacturer's machine.

By October 2012, two manufacturers had completed the field testing with a combined 21 days of personal breathing zone sampling at 11 different highway construction sites [NIOSH 2013b,c]. The 42 personal breathing zone air sampling results (21 days with 2 workers per day) were all below the NIOSH REL of 0.05 mg/m^3 for respirable crystalline silica, ranging from below the limit of detection up to 0.024 mg/m^3 for the milling machines with well-designed dust controls.

Silica in Construction – Asphalt Pavement Milling

Many construction tasks have been associated with overexposure to crystalline silica [Rappaport et al. 2003]. Among these tasks are tuck pointing, concrete sawing, concrete grinding, concrete breaking, jackhammering, and abrasive blasting [NIOSH 2000; Thorpe et al. 1999; Akbar-Khanzadeh and Brillhart 2002; Glindmeyer and Hammad 1988]. Road milling has also been shown to result in overexposures to respirable crystalline silica [Linch 2002; Rappaport et al. 2003; Valiante et al. 2004]. However, the three road-milling studies do not provide enough information about the operating parameters and engineering controls present on the milling machines to determine if the overexposures were due to a lack of effective controls or poor maintenance of the machines.

A variety of machinery is employed in asphalt pavement recycling, including cold-planers, heater-planers, cold-millers, and heater-scarifiers [Public Works 1995]. Cold-milling of asphalt pavement is the focus of this document. Cold-milling, which uses a toothed, rotating cutter drum to grind and remove the pavement to be recycled, is primarily used to remove surface deterioration on both petroleum-asphalt aggregate (the focus of this document) and Portland-cement concrete road surfaces [Public Works 1995]. Key components of a typical half-lane and larger asphalt pavement milling machine are shown in Figure 1.

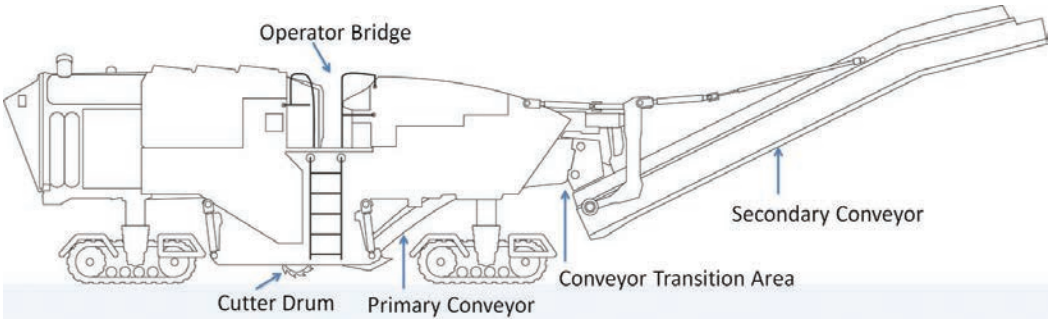


Figure 1. Half-lane and larger asphalt pavement milling machine (Illustration by NIOSH)

Approximately 367,000 U.S. workers are employed in highway, street, and bridge construction [U.S. Census Bureau 2007]. A number of these workers use cold-milling machines or work in close proximity to these machines. Dust generated from the cold-milling machines often contains respirable crystalline silica. This respirable dust can be transported by air currents to worker breathing zones near the milling machine.

Guidelines for Dust Control on Asphalt Milling Machines

The guidelines in this document for half-lane and larger asphalt milling machines are the result of the collaborative work of labor, industry, and government partners. The participating cold-milling machine manufacturers entered into the research project with the Partnership to reduce respirable crystalline silica exposures and improve working conditions in highway construction. The following recommendations are the result of this collaborative research partnership; ventilation recommendations are provided first, followed by guidelines for water-spray dust suppression.

Ventilation Control Guidelines for Pavement-Milling Machines

NIOSH recommends the use of ventilation controls to control worker exposures to respirable crystalline silica during asphalt pavement milling in highway construction. The ventilation controls used in combination with water-sprays that cool the cutting teeth can consistently reduce exposures below the NIOSH REL of 0.05 mg/m^3 [NIOSH 2013b,c].

The ventilation controls designed to reduce respirable crystalline silica exposures on asphalt pavement milling machines should be evaluated in a controlled setting (i.e., shop or warehouse) in which their efficiency can be measured using smoke and tracer gas tests. The visible smoke test should be used as a qualitative test to visualize air-flow patterns around the milling machine drum housing and conveyors to assist in setting up for tracer gas tests. The tracer gas test should be used as a quantitative measure of the air-flow rate and capture efficiency of the ventilation system designed to control the release of respirable crystalline silica into the workers' breathing zones. Laboratory tracer gas and smoke release test procedures to evaluate ventilation controls on asphalt pavement milling machines are provided in Appendix A.

When designing a ventilation control on a milling machine, the designer should consider the following:

- extent of enclosure around the drum housing and conveyors
- hood and duct design
- required airflow capacity
- durability and noise of the duct and fan
- measures to prevent clogging of the ventilation control

The drum housing and conveyors on the milling machine should be designed to provide maximum enclosure around the cutter drum and conveyor belts. An ideal design would have a drum housing and conveyor system that is 100% enclosed with only an opening at the top of the secondary conveyor where recycled asphalt pavement is transferred to the dump truck.

Eliminating the open area in the design of these enclosures will help maintain negative air pressure necessary for containment of silica dust. Research conducted by the Partnership has shown that a well-enclosed drum housing and conveyor system can reduce workers' dust exposure with as little as 900 actual cubic feet per minute (acfm) of air drawn from the primary conveyor and drum housing areas. The minimum flow rate of 900 acfm was determined to be effective in several tracer gas studies [NIOSH 2011c; 2012; 2013a].

Providing negative air pressure within an enclosure is a common ventilation technique for preventing a contaminant from escaping into surrounding areas. To contain silica dust, it is important to maintain negative air pressure in the drum housing of the milling machine. Operation of numerous water-spray nozzles within the drum housing can put the housing under positive air pressure. If exhaust ventilation is not used, dust-laden air can be forced out through gaps in the drum housing and contaminate the ambient surroundings, including the breathing zones of adjacent workers. The rotating drum and moving asphalt material may also generate air flows that can lead to dust escape from the drum housing. It is therefore important that the amount of open area through which leaks may occur from the drum housing be minimized to the extent compatible with the work process. The ventilation control should remove enough air to overcome any positive pressure and set a negative pressure differential between the drum housing and the surrounding air. *Industrial Ventilation: A Manual of Recommended Practice for Design* recommends a negative pressure differential of 0.04 +/- 0.02 inches of water as a good performance standard for containment of most industrial processes [ACGIH 2013b].

Typical ventilation controls designed to reduce dust exposures in a plant or on a piece of equipment consist of a hood, fan, ductwork, and dust collector [ACGIH 2013b]. If the ventilation control on a milling machine does not include a dust collector, the outlet from the system should discharge at high velocity in an upward orientation and be located near the top of the secondary conveyor or in another location away from workers. Milling machines without a dust collector should also be designed to allow the operator to temporarily turn the ventilation control off when milling into the wind. However, a feature should be in place to automatically turn the ventilation control back on when it has been off for longer than 60 minutes. Water-spray systems for cooling cutting teeth should remain on regardless of wind direction.

The ventilation control should be designed to have a minimum transport air velocity through the duct to prevent dust from settling and plugging the flow. The minimum duct design velocity recommended by American Conference of Governmental Industrial Hygienists (ACGIH) is 4,500 ft/min for heavy or moist dust [ACGIH 2013b].

Duct transport velocities alone may not be enough to prevent plugging. It is also important to design a gradual transition of air velocities between the duct and the take-off point of the dust control hood so that larger particles do not enter and clog the duct or damage the fan. The fan could move air from the drum housing or primary conveyor through a hood or settling box to gradually transition the air velocity from low to high before air enters the duct. The design intent of the ventilation control should be to at least control respirable size silica particles. It may also be prudent to design the ventilation control to capture inhalable size dust particles up to 100 μm in diameter. Particles less than 10 μm are capable of entering the gas exchange region of the lungs and triggering silicosis and cancer [Plog 2002].

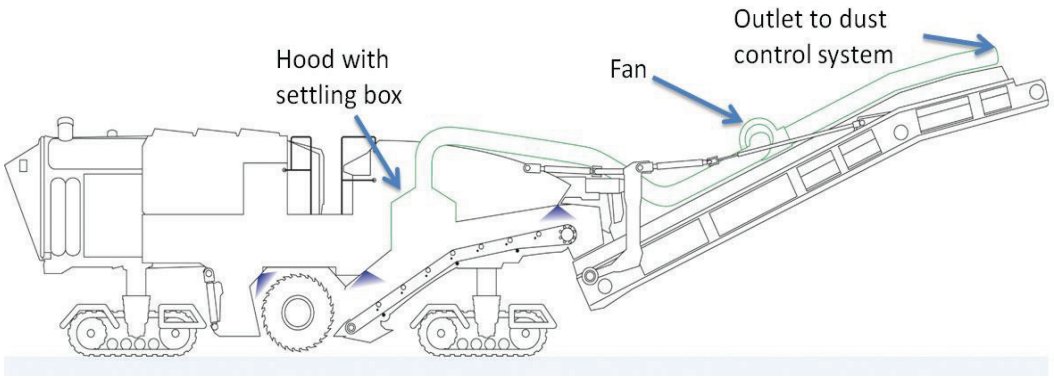


Figure 2. Hood and settling box on a milling machine (Illustration by NIOSH)

Aerosol scientists have developed a spreadsheet that can be used to calculate settling velocity based on dust particle size and other factors [Baron and Willeke 2001]. Milling machine manufacturers should use the information about particle settling velocity to design a gradual transition of air velocities that prevent larger particles from entering the duct. A possible design would be to orient a hood or settling box so that air is pulled vertically from the drum housing or conveyor. Figure 2 shows a square settling box designed to settle out larger particles by gradually transitioning air velocities from low to high before dust enters the duct.

Even with a settling box or hood to provide a gradual transition of air velocities, the fan and ductwork should be designed to withstand a dusty airstream. The durability of the fan and duct work should be evaluated during the design and testing phase of the ventilation control. Any deficiencies in durability of the fan and duct work or clogging should be corrected before implementing the ventilation control across a fleet of milling machines.

Water Spray Application Guidelines for Asphalt Milling Machines

The Partnership conducted considerable research to develop best practice recommendations for water-spray dust suppression on asphalt milling machines. NIOSH recommends using best practice water-spray systems on asphalt milling machines that do not have ventilation controls. Water-spray systems can provide an effective and economical method to control dust in many types of industrial applications. Although the use of water sprays is a very simple technique, a number of factors should be evaluated to determine the most effective design for their use on asphalt pavement milling machines. Two effective dust control methods utilizing water sprays that are readily applicable to asphalt pavement milling operations are dust prevention and suppression techniques. Dust prevention is achieved by spraying water onto the pavement being milled in order to prevent dust from being liberated/generated and becoming airborne. The second method, dust suppression, involves knocking down airborne dust by spraying the dust cloud and causing the particles to collide, agglomerate, and fall from the air. A common and effective practice is to use a combination of both of these wet methods in the overall dust control plan. Since water is used on all asphalt pavement-milling machines to cool the cutter bits, it is logical to also use this water for respirable dust control.

The primary dust prevention technique is to wet road material (asphalt pavement) during milling, thus causing the material, fines, and dust particles to increase in weight. This in turn, decreases the ability for these dust particles to become airborne. As groups of particles become heavier, it becomes more difficult for the surrounding air to carry them off. The keys to effective wet spray dust control are (1) water pressure and water flow rate, (2) the proper application of moisture, (3) the careful design of nozzle location and orientation, (4) the control of droplet size, (5) the selection of the best spray pattern and spray nozzle type, and (6) the proper maintenance of nozzles and water spray application equipment [NIOSH 2010]. To the extent possible, the tanks, pumps, lines and nozzles in a dust suppression system should be made of non-corrosive or corrosion resistant materials. One last consideration is to have a properly designed liquid filtration system prior to the water application to minimize contaminants in the system and, ultimately, to prevent clogged water spray nozzles. *NIOSH Best Practices for Dust Control in Metal/Nonmetal Mining* provides more detailed information about spray nozzle selection and settings that have been effective in dust control for underground mining machines that share similar technology with cutter drums on milling machines [NIOSH 2010].

During the initial separation from the road bed, adequate wetting is extremely important to ensure that dust particles stay attached to the broken asphalt pavement material. Uniform wetting is also an important issue for an effective system. Dust reductions can be achieved by spraying the milled material with water while using the cutter drum to mechanically mix the asphalt pavement and water together to achieve uniform wetting [NIOSH 2011d]. The water sprays can cool the cutting bits on the milling drum and also provide dust suppression. Figure 3 illustrates water nozzles spraying the bits on a cutter drum. To conserve water, the water sprays should only be activated during milling. It is also recommended that manufacturers install water flow meters and pressure gauges on each machine to allow for visual determination that the water spray system is operating within the designed operating range.



Figure 3. Water application at drum to cool cutting bits. The water is then mixed with asphalt pavement material as it is removed by the cutter bits (Photo by NIOSH)

Water Spray Application—Case Studies

In cooperation with NAPA, milling contractors, and equipment manufacturers, NIOSH evaluated the effectiveness of changes in water spray applications on multiple pavement milling job sites. During these surveys, respirable dust levels were measured at multiple locations around the milling machine to monitor dust liberation from the cutter housing and lower conveyor transfer locations. Descriptions of the sampling procedures used during these surveys and specific test conditions at each site are provided in technical reports referenced with each case study. Not all water spray modifications were successful, but multiple surveys did provide examples of measurable reductions in dust liberation. Several case studies are summarized below to illustrate the modifications and changes in water spray application and their measured benefits.

Case Study A

On this milling machine, baseline dust levels were determined with the water spray system operated at levels routinely used during milling. The system was equipped with 18 flat-fan sprays (i.e. nozzles) located behind the cutter drum, five flat-fan sprays at the front of the cutter drum, and several sprays at the transition from the primary conveyor. The water pump on the milling machine could be adjusted and had a rated output up to 15 gallons per minute (gpm) at 200 pounds per square inch (psi). However, the water spray system levels routinely used during milling were 6 gpm at 35 psi.

Adjustments were made to the pump to increase the water pressure from 35 psi to 70 psi with a corresponding increase in water flow from 6 gpm to 12 gpm. Figure 4 shows a typical water flow meter and pressure gauge that were installed and used to quantify water application changes during testing. Measurements from seven area dust sampling locations around the machine resulted in a 50% reduction (from a geometric mean of 1 mg/m^3 to 0.5 mg/m^3) in respirable dust by simply adjusting the water pump from the operator's deck from 35 to 70 psi, without any change or addition to the existing spray nozzles [NIOSH 2007a]. Personal breathing zone air sampling results for respirable crystalline silica ranged from less than the limit of detection to 0.170 mg/m^3 for the trial time sampled [NIOSH 2007a].

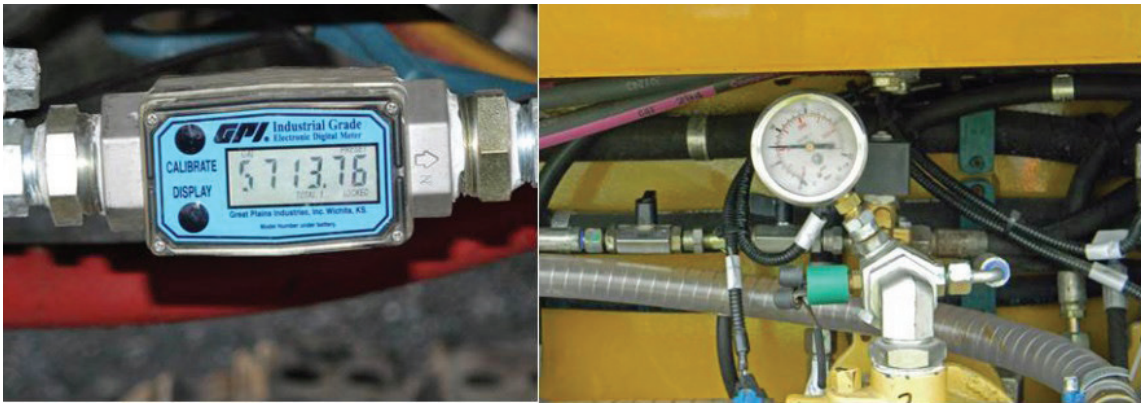


Figure 4. Water flow meter (left) and pressure gauge (right) used during pavement milling machine studies (Photo by NIOSH)

Case Study B

Similar to Case Study A, the water flow to this milling machine could be adjusted by the operator by changing the output from the water pump. During this survey, increases in water flow from 12.5 gpm to 18.8 gpm resulted in respirable dust reductions of approximately 41% (from a geometric mean of 0.328 mg/m^3 to 0.193 mg/m^3) at 10 key area sampling locations around the machine [NIOSH 2009b]. Once again, these dust reductions were achieved without making changes to the existing water spray system other than increasing the water pump output.

Case Study C

During this evaluation, the baseline water spray system consisted of a total of 16 water spray nozzles located at the front and rear of the cutter housing and at the discharge to the primary conveyor. Total water flow for these nozzles was 22.4 gpm at 20 psi. For the modified spray system, a total of 14 nozzles were used, which supplied 21.4 gpm at 20 psi. In this case study, water quantity was not increased, but changes to the distribution of the water, such as spray nozzle location and orientation, improved dust control. For example, nozzles were relocated in the front of the cutter housing to discharge water in the opposite direction of material flow, the upper front-housing sprays were eliminated, and sprays were added to the primary conveyor transition. Respirable dust levels measured for the modified spray system at six key area sampling locations on the machine resulted in an average dust reduction of approximately 43% (from a geometric mean of 0.72 mg/m^3 to 0.41 mg/m^3) when compared to baseline dust levels [NIOSH 2011d].

The Partnership determined that Case Study C was the most effective dust control configuration evaluated on a new asphalt milling machine and engineering drawings are provided in Figures 5-7. However, this recommendation should not limit manufacturers from implementing future improvements in water-spray dust control as new technologies, configurations, or inventions are evaluated.

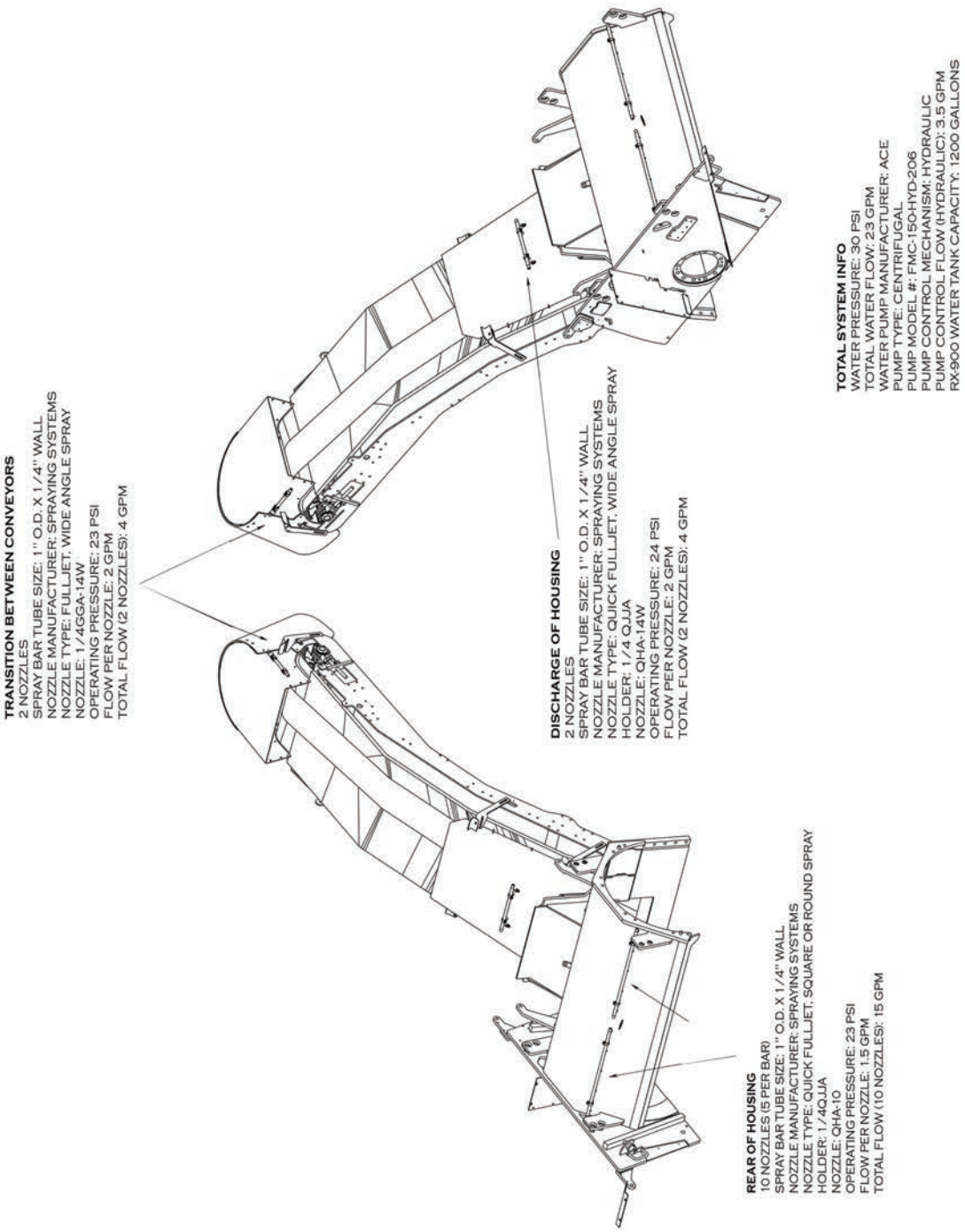


Figure 5: Water-spray details overview

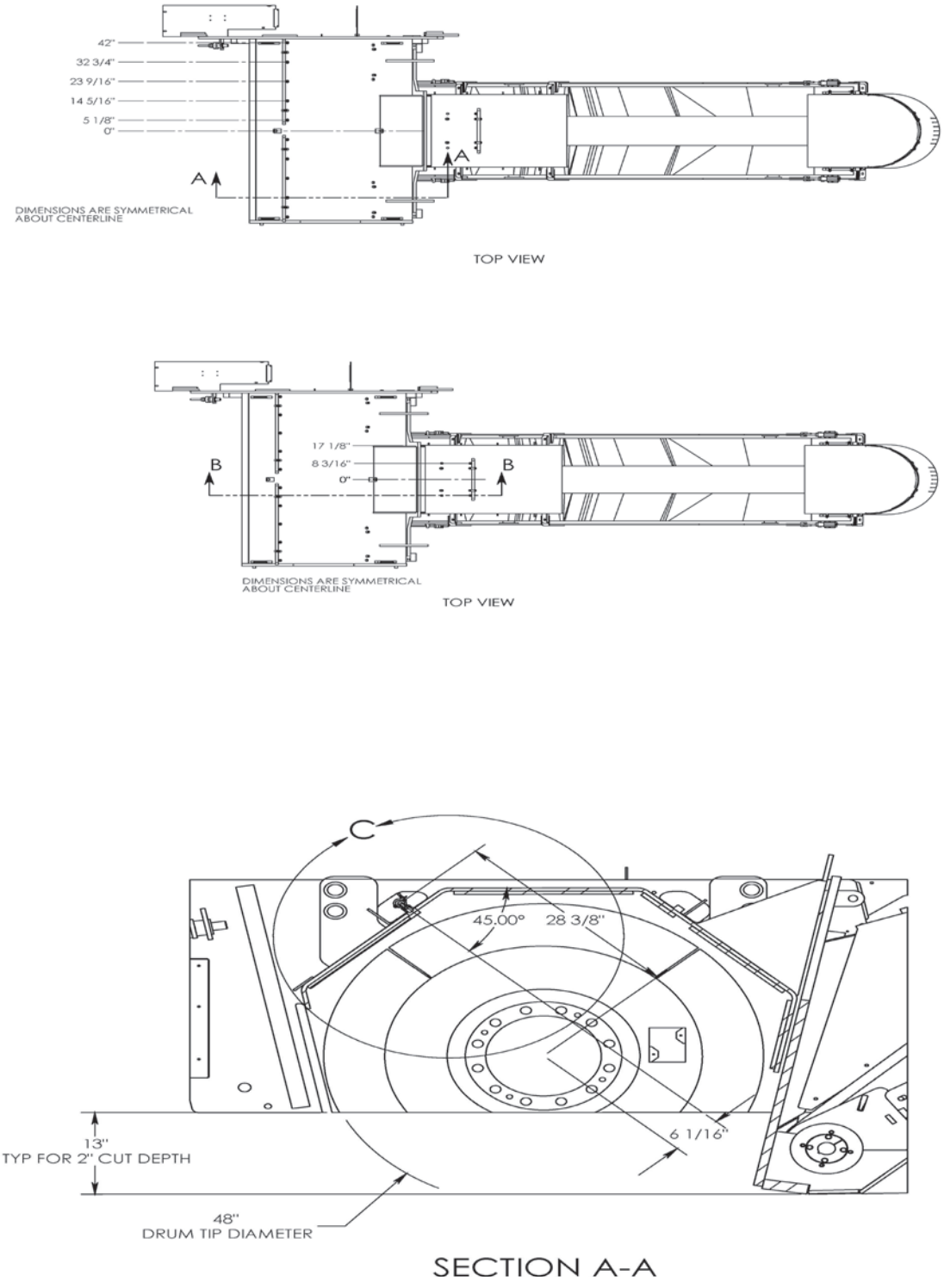
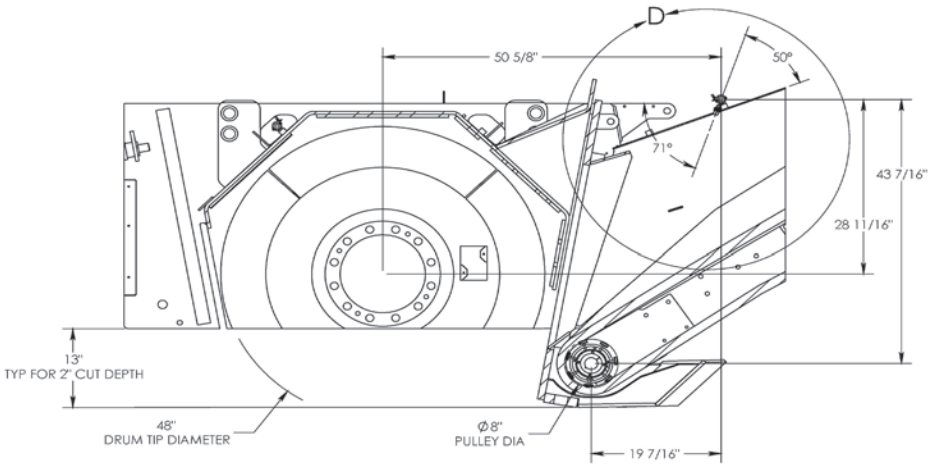
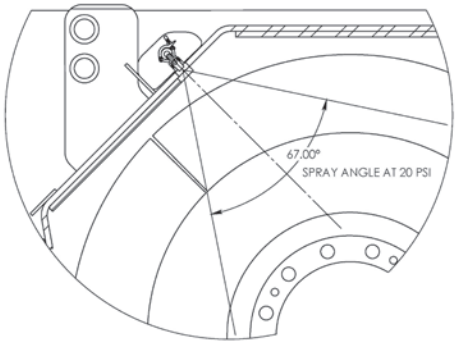


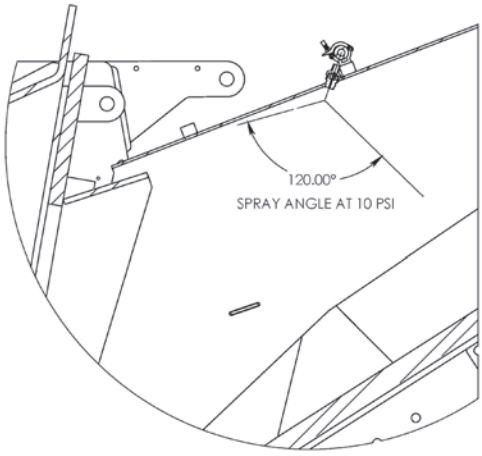
Figure 6: Water-spray details drum housing



SECTION B-B

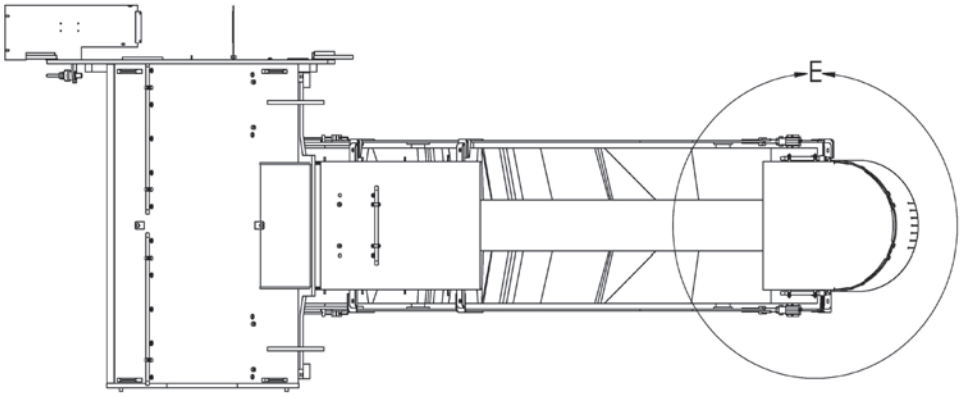


DETAIL C

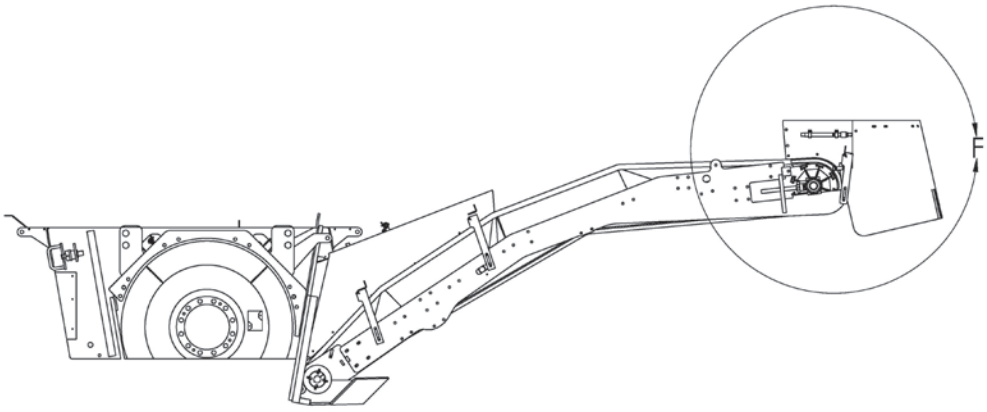


DETAIL D

Figure 6 (cont.): Water-spray details drum housing

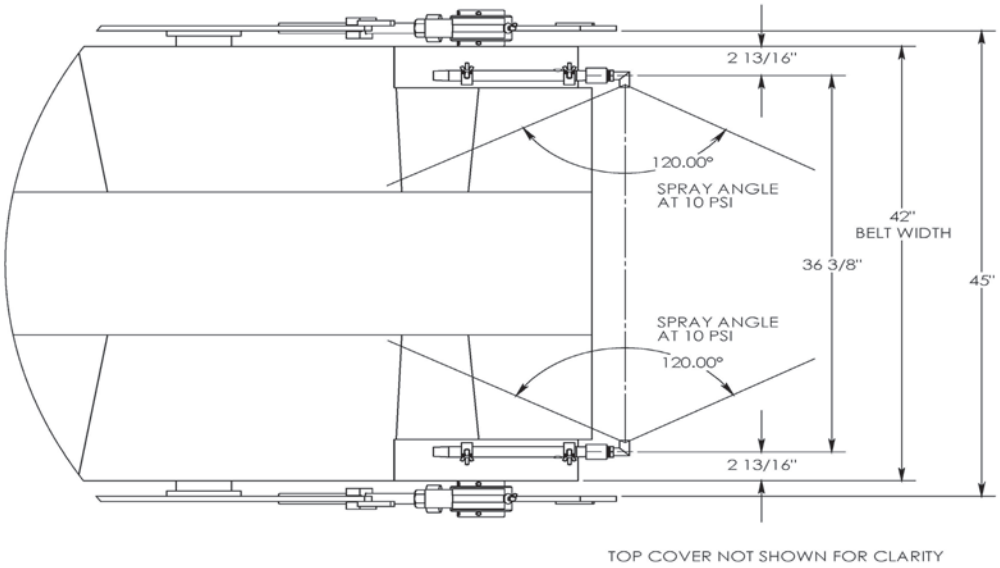


TOP VIEW

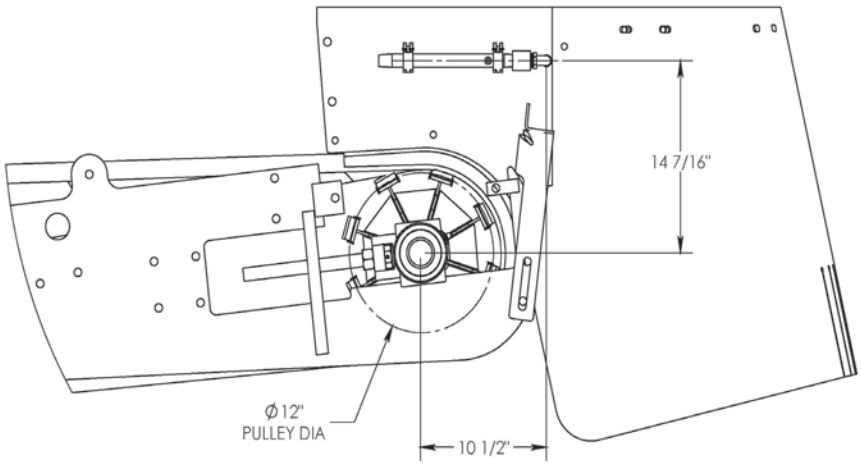


SIDE VIEW

Figure 7: Water-spray details conveyor transition



DETAIL E



DETAIL F

Figure 7 (cont.): Water-spray details conveyor

Wet Drum Studies

In addition to multiple evaluations of water spray configurations, NIOSH and the Partnership also evaluated the utilization of wet drum technology. With the wet drum, water is supplied to the inside of the cutting drum and uses gravity to release water through nozzles in the drum housing in an effort to prevent the release of dust in the cutting area. The dust levels obtained with the wet drum were compared to dust levels generated by the baseline water spray system, which consisted of sprays located along the back side of the cutting drum and sprays at the primary and secondary conveyor transfer locations. The wet drum technology was evaluated at two different milling sites. At the first test site, utilization of the wet drum resulted in a 32% reduction at the lower sampling locations and a dust reduction of 59% at the operator location [NIOSH 2011c]. At the second test site, the wet drum resulted in average dust reductions of 37% at the lower sampling locations and 27% at the operator location [NIOSH 2013d]. However, due to wide performance variability, the reductions at the second site were not statistically significant. Personal breathing zone air sampling results for respirable crystalline silica ranged from 0.028 mg/m^3 to 0.085 mg/m^3 at the second wet drum test site [NIOSH 2013d].

One problem observed with the operation of the wet drum technology was inconsistent water distribution from the drum. On level roadways, water was discharged from the drum evenly across the entire milled strip. However, if the roadway was not level across its width, water would discharge from the low side of the cutter drum but the roadway on the high side of the drum would be dry. Baffles added inside the drum to ensure uniform water distribution to all sections of the drum might potentially improve the performance of the wet drum technology.

NIOSH Recommended Best Practice Test Guidance for Dust Controls on Milling Machines

- NIOSH recommend that each milling machine manufacturer design and install ventilation controls with a minimum indoor capture efficiency of at least 90% (as determined by the tracer gas method described in Appendix A) on all new half-lane and larger cold-milling machines. Each milling machine manufacturer should conduct the tracer gas testing of ventilation controls for a representative design of a half-lane and full-lane new milling machine. Manufacturers should consider re-testing future milling machine designs if appreciable changes are made that could negatively impact the effectiveness of the ventilation control.
- Manufacturers should install an air-flow indicating device on each milling machine so that the ventilation control flow rate can be monitored. The device should allow for visual determination that the ventilation control is operating within the designed operating range. Each manufacturer should attach a plate to the milling machine showing a simple diagram of the ventilation control and a list of field-maintenance procedures.
- NIOSH recommends that each manufacturer should use the procedures in Appendix B to conduct field testing of a representative design of a half-lane and full-lane new milling machine equipped with ventilation controls. The purpose of the field testing is to verify that the ventilation controls continue to protect workers during asphalt pavement milling.

- The ventilation and water-spray dust controls described in this document represent the current knowledge of best practices developed by the Partnership for controlling dust during asphalt pavement milling. This document should not limit manufacturers from implementing future improvements in dust control as new technologies or inventions become available. Manufacturers can use the procedures in Appendix B to evaluate current and future dust controls. Appendix D provides instructions on how to use the spreadsheet that accompanies the statistical procedures explained in Appendix B.
- Milling machine contractors should implement a water-spray and ventilation control system daily checklist to verify that the dust controls are being operated according to the manufacturer instructions. An example daily checklist is provided in Appendix E.

Recommendations for Operation and Maintenance of Dust Control Systems

Cold-milling machine manufacturers should provide an operator's manual that contains a maintenance schedule for the water-spray or ventilation controls. Each operator's manual should also include detailed sketches, performance criteria, and troubleshooting instructions for equipment owners to use in their periodic inspection of these controls. Owners of cold-milling machines equipped with these dust control systems are advised to inspect and maintain the systems according to the manufacturer's recommendations and to record the dates of periodic inspections in the operator's manual.

Training

Equipment manufacturers should provide training manuals to equipment owners in the operation, maintenance, and repair of their dust control systems for cold-milling machines. Equipment mechanics trained in the maintenance and repair of cold-milling machines may require additional training to be aware of and perform these services on the dust control systems. Milling machine operators should also be trained on these controls and be provided with the desired operating specifications (water quantity, water pressure, air velocity) that must be maintained to be effective. Also, milling machine operators should be trained to temporarily disable the ventilation control system when milling directly into the wind.

Additionally, NAPA has published a Best Practice Bulletin 1/12, *Operational Guidance for Water Systems During Milling Operations,* which is reprinted in Appendix F of this document with permission from NAPA. The NAPA Best Practices Bulletin 1/12 is free of charge and provides additional information about maintaining water-spray systems during milling operations.

Silica and Health Effects

Silica (SiO_2), an oxide of silicon, occurs in nature as a mineral in quartz, chalcedony, chert, flint, opal, diatomaceous earth, and sandstone [USGS 1914]. "Crystalline" refers to the orientation of the SiO_2 molecules in a fixed pattern as opposed to a nonperiodic, random molecular arrangement

defined as amorphous [NIOSH 1974]. The three most common forms of crystalline silica encountered in construction and mining are quartz, tridymite, and cristobalite. NIOSH defines respirable crystalline silica as

> "*that portion of airborne crystalline silica that is capable of entering the gas-exchange regions of the lungs if inhaled; by convention, a particle-size-selective fraction of the total airborne dust; includes particles with aerodynamic diameters less than approximately 10 μm and has a 50% deposition efficiency for particles with an aerodynamic diameter of approximately 4 μm*" [NIOSH 2002].

NIOSH estimates that at least 1.7 million American workers are exposed to respirable crystalline silica and that many of those workers' exposures exceed current occupational exposure limits [NIOSH 2002]. Inhalation of respirable crystalline silica can cause silicosis, a debilitating and potentially fatal lung disease. Silica exposure has also been associated with lung cancer, chronic obstructive pulmonary disease, renal disease, and other adverse health outcomes [NIOSH 2002]. During the period from 1990 through 1999, at least one-third of decedents with silicosis had worked in construction or mining [NIOSH 2007b].

Occupational Exposure Limits

In the United States, occupational exposure limits (OELs) have been established by federal agencies, professional organizations, state and local governments, and other entities. The U.S. Department of Labor OSHA permissible exposure limits (PELs) [29 CFR 1910.1000 (2003)] are OELs that are legally enforceable in covered workplaces under the Occupational Safety and Health Act. NIOSH issues RELs that are based on a critical review of the scientific and technical information available on the prevalence of health effects, the existence of safety and health risks, and the adequacy of methods to identify and control hazards [NIOSH 1992]. Other OELs that are commonly used and cited in the United States include the Threshold Limit Values (TLVs) recommended by ACGIH [ACGIH 2013a]. ACGIH TLVs are considered voluntary guidelines for use by industrial hygienists and others trained in this discipline "to assist in the control of health hazards." Workplace environmental exposure levels (WEELs) are recommended OELs developed by the American Industrial Hygiene Association (AIHA). WEELs have been established for some chemicals "when no other legal or authoritative limits exist" [AIHA 2007].

OSHA requires an employer to furnish employees a place of employment that is free from recognized hazards that are causing or are likely to cause death or serious physical harm (Occupational Safety and Health Act of 1970, Public Law 91–596, sec. 5(a)(1)). Thus, employers are required to comply with OSHA PELs. NIOSH investigators also encourage employers to consider other OELs in making risk assessment and risk management decisions to best protect the health of their employees. NIOSH investigators encourage the use of the traditional hierarchy-of-controls approach to eliminating or minimizing identified workplace hazards. This approach includes, in preferential order, the use of (1) substitution or elimination of the hazardous agent, (2) engineering controls (e.g., local exhaust ventilation, process enclosure) (3) administrative controls (e.g., limiting time of exposure, employee training, work practice changes, medical surveillance),